SUSTAINABILITY

Scaling Digital Infrastructure in Manufacturing:
Real-Time Telemetry for Carbon Reduction and Dynamic LCA

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Recap: Challenge → potential → solution

Electricity: Root cause of manufacturing emissions

Challenges in quantifying demand…

- Life cycle assessment (LCA) data collection and modeling can be **time-consuming and costly**
- LCA background data **lags behind current industry practices,** and requires process-level activity data
- **Measuring highly granular activity data is complex** (temporally, geospatially, technologically)

What matters most in manufacturing…

- Electricity tends to be **greatest contributor to gate-to-gate emissions (>90%)**
- Across all manufacturers, annual electricity usage likely in the **tens of terawatt-hours (TWh) globally**
- Electricity consumption has tremendous **environmental and financial impact**

Untapped potential of **manufacturing data**

Sustainability

Without real-time data on industrial activity, we can't pinpoint which manufacturing processes contribute the most to environmental burden.

Manufacturing operations

Without advanced analytics, factories are missing the opportunities to significantly improve operational efficiencies.

Solution: Digital infrastructure in manufacturing

Deployment of digital infrastructure to enable direct measurement of energy consumption to enable key use cases of (1) generating product-level carbon disclosures for comparability and (2) providing energy insights.

(1) **Product-level inventories:** Data output packaged to align with life cycle inventory (LCI) standard as per ISO 14040/44/48 and ILCD.

(2) **Energy insights:** pinpoint energy consumption through direct measurement, giving clear insight into reduction opportunities and clean energy strategies. We can also apply data science to further analyze this data, providing deeper insights and driving meaningful change.

2025 to 2026: Idea to impact

2025

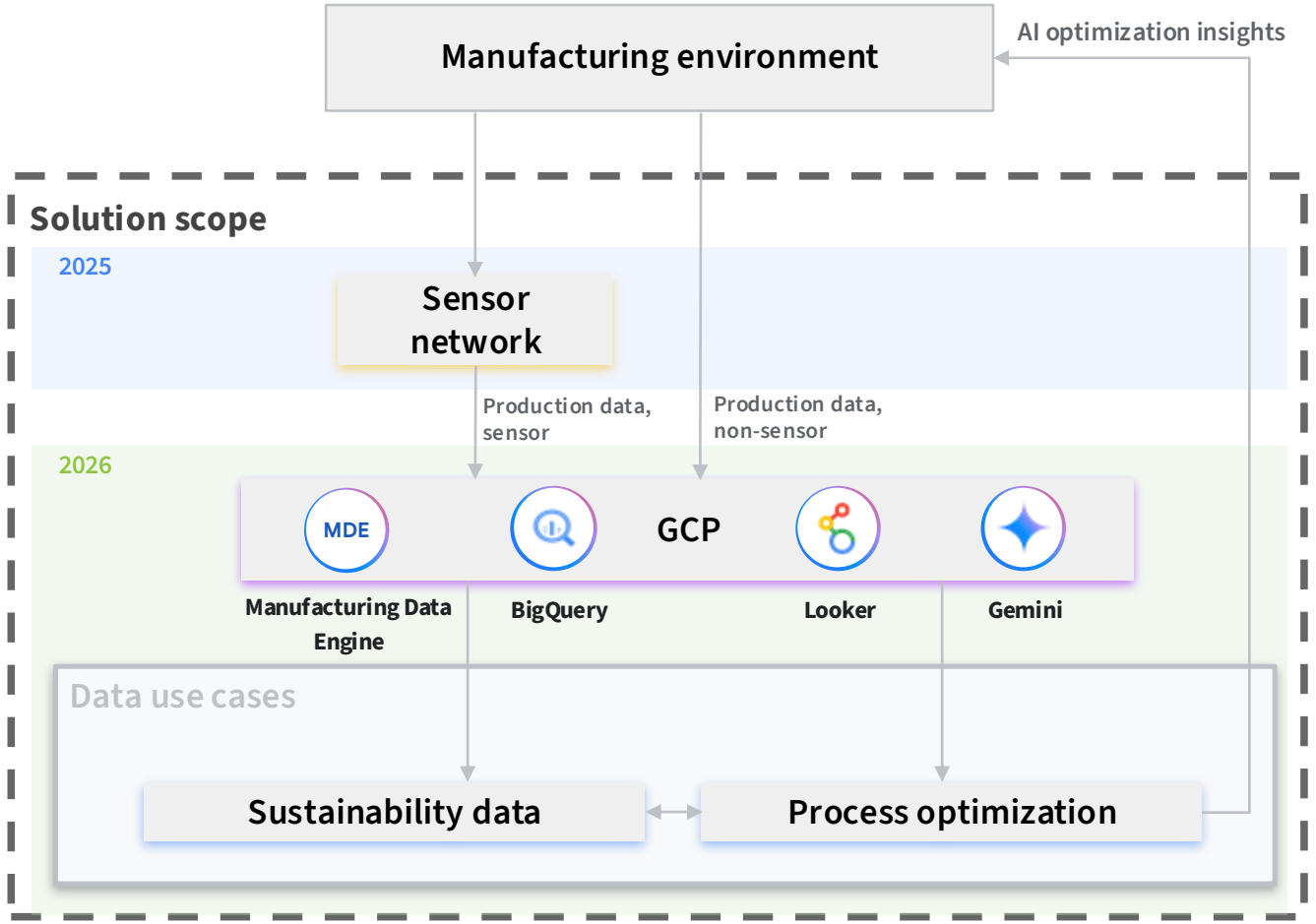
1 | At this time last year, we had developed the **physical data infrastructure** to measure and transmit data.

Since then, we've implemented a **unified, enterprise grade architecture.**

2026

2 | We utilized **Google Cloud's Manufacturing Data Engine**, or MDE, as the backbone. This allows us to unify disparate physical sources - everything from smart electricity meters to assembly equipment sensors into a high-performance time-series database.

3 | This approach unlocks two key use cases, **generating life cycle inventories** and **unlocking AI-enabled optimization.**



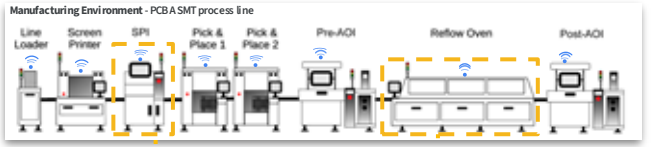
Real data in real-time: Unified OT/IT architecture

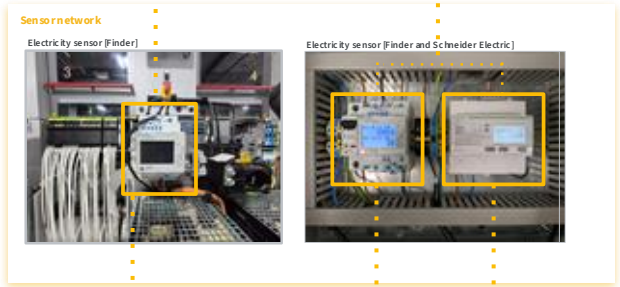
1 | OT floor pipeline

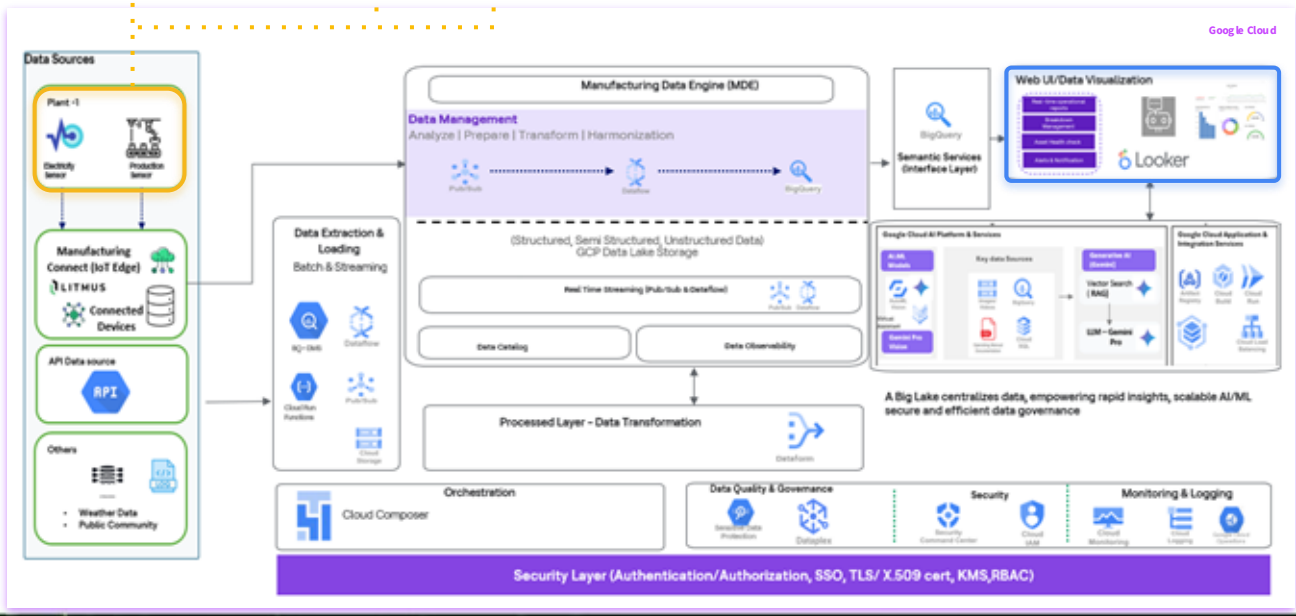
We built a unified OT/IT data pipeline. At the factory floor, physical OT devices - from electrical sub-meters to equipment telemetry - transmit data via LoRa gateways and edge connectors.

2 | IT Contextualization

This data feeds directly into GCP's MDE, which breaks down traditional factory silos by contextualizing raw, time-series data with production context - linking kilowatt-hours directly to specific unit serial numbers, work orders, and timestamped process steps. This creates an auditable, high-resolution data pipeline that serves both factory operators and sustainability teams.







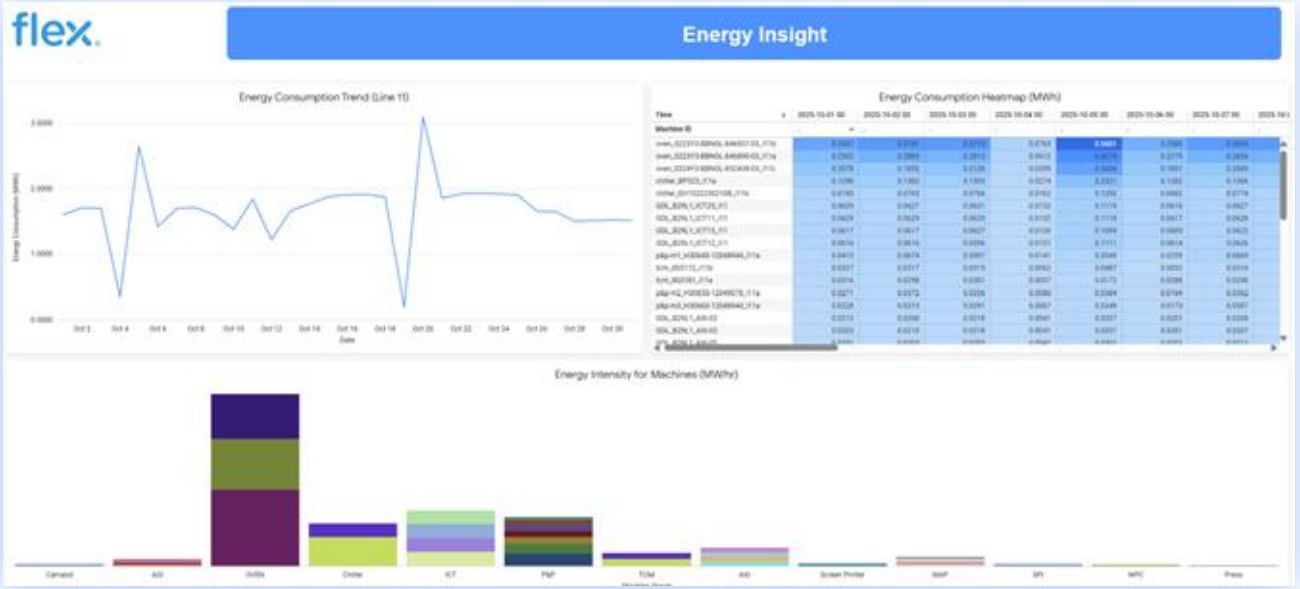
Real data in real-time: Empirical power profiling

3 | Programmatic isolation of energy consumption

Machine-level tracking: High-resolution energy monitoring across the ML platform surface-mount technology (SMT) line, seamlessly merging sensor data with operations for a single, unified view of energy consumption

Cross-functional insights: Comprehensive usage mapping for Manufacturing and Testing, pinpointing exactly when and where power is consumed

Intelligent monitoring and smart alerts: Real-time insights to flag anomaly-based notifications and trigger predictive maintenance workflows.



Google Real

Energy Insight

just now

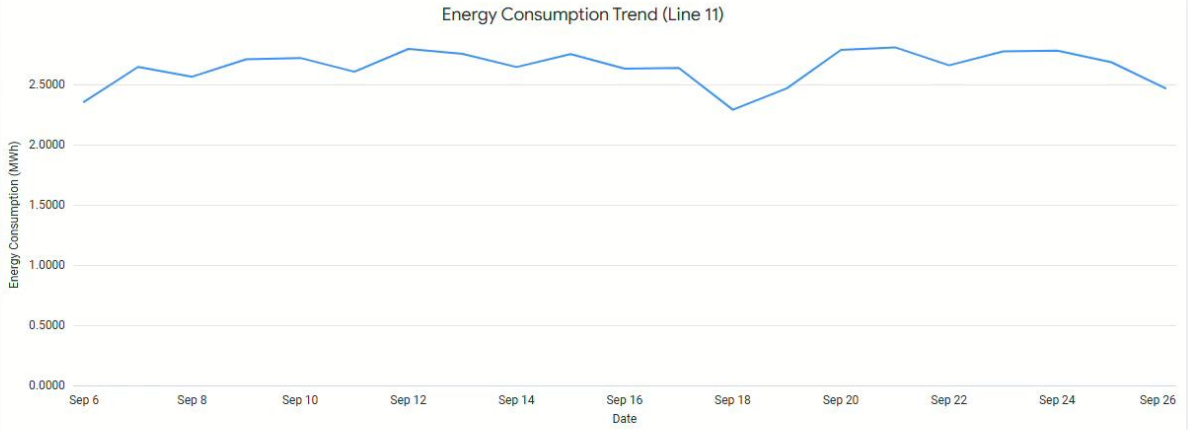
Select Date Range * | SMT/IST | Machine Type

last 30 days | SMT | is any value

flex.

Energy Insight

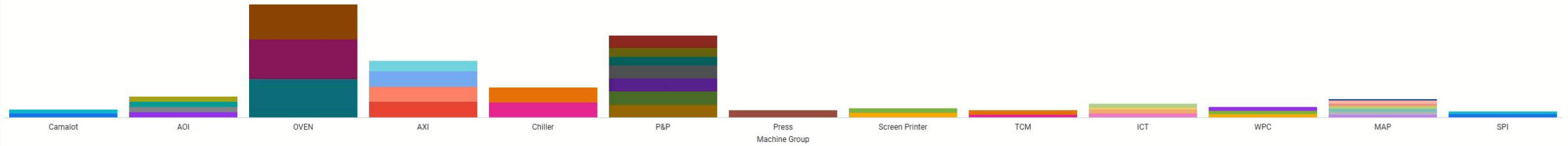
Energy Consumption Trend (Line 11)



Energy Consumption Heatmap (MWh)

Time	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...	2026-09-...
Machine ID	.	.	.	.	.	.	.	.	.	.	.	.	.
oven_0223Y3-BBNGL-846890-03_I11a	0.2882	0.2837	0.2873	0.2594	0.2529	0.2853	0.2852	0.2816	0.2877	0.2762	0.2806	0.2811	0.22
oven_0223Y3-BBNGL-846857-03_I11b	0.2782	0.2811	0.2857	0.2872	0.2848	0.2838	0.2763	0.2221	ø	0.2860	0.2840	0.2848	0.28
chiller_BP523_I11a	0.1122	0.1138	0.1160	0.1155	0.0990	0.1003	0.1128	0.1002	0.1134	0.1113	0.1047	0.1103	0.11
GDL_B29L1_AXI-03	0.1091	0.1341	0.0799	0.1227	0.0961	0.1325	0.1058	0.1017	ø	0.1050	0.0654	0.1008	0.10
GDL_B29L1_AXI-02	0.1068	ø	0.1110	0.1256	0.1273	0.1116	0.1293	ø	0.0897	0.1114	0.1249	0.1058	0.12
p&p-m2_H3083S-12049078_I11a	0.1067	0.0958	0.0867	0.0751	0.1072	0.1054	0.0786	ø	0.1053	0.1038	0.0976	ø	0.10
p&p-m2_H3097S-12049148_I11b	0.1044	0.0886	0.1014	0.1020	0.0880	0.0944	0.0859	0.0877	0.0867	0.0937	0.0972	0.1010	0.10
p&p-m1_H3064S-12048944_I11a	0.1040	0.0911	0.0808	0.0959	ø	0.0913	0.1071	0.1039	0.1040	0.1011	0.1011	0.1052	0.10
p&p-m1_H3111S-12049104_I11b	0.1022	0.0956	0.0955	0.0715	0.0953	0.1026	0.0890	0.0865	0.0928	0.1022	0.1047	0.1042	0.09
chiller_EH1522230210B_I11b	0.1002	0.1133	0.1110	0.1119	0.1105	0.1123	0.0934	0.1114	0.0839	ø	0.0921	0.1129	0.11
GDL_B29L1_AXI-01	0.0960	0.1221	0.0934	0.1229	0.1287	0.0840	0.1035	0.1298	0.1366	0.1336	0.1238	0.1286	
p&p-m3_H3066S-12048944_I11a	0.0703	0.0627	0.0569	0.0669	0.0619	0.0695	0.0709	0.0565	0.0719	0.0690	0.0643	0.0703	0.05
p&p-m3_H3084S-12049078_I11b	0.0687	0.0614	0.0719	0.0711	0.0564	0.0695	0.0711	0.0414	0.0703	0.0662	0.0540	0.0708	0.06
GDL_B29L1_AXI-05	0.0674	ø	0.0777	0.0715	0.0612	0.0488	0.0985	0.0934	0.0842	0.0979	0.0998	0.0949	0.08
p&p-sx2_R1838G-12049148_I11b	0.0624	0.0982	0.1038	0.0811	0.1010	0.0955	0.0941	0.0943	0.0808	0.0651	0.0893	0.0833	0.09
GDL_B29L1_AOI-02	0.0454	0.0409	0.0463	0.0449	0.0456	0.0445	0.0448	0.0385	ø	0.0355	0.0434	0.0445	0.03
pressfit_2301269_I11b	0.0452	0.0542	0.0469	0.0527	0.0588	0.0475	0.0643	0.0551	0.0402	0.0153	0.0481	0.0521	0.04

Energy Intensity for Machines (MWh)



OCP GLOBAL SUMMIT | 2026

Scaling Innovation for the AI Era.

Data use case: Move towards dynamic LCA

System architecture for
on-demand life cycle inventories
from raw factory data to *standardized output*

System requirements |
Core framework

1. Dynamic generation + standardization
Compile inventories (ILCD, JSON-LD, EcoSpold2) from live factory data.

2. Secure data interoperability
Enable robust, bi-directional LCI data exchange with partners and suppliers.

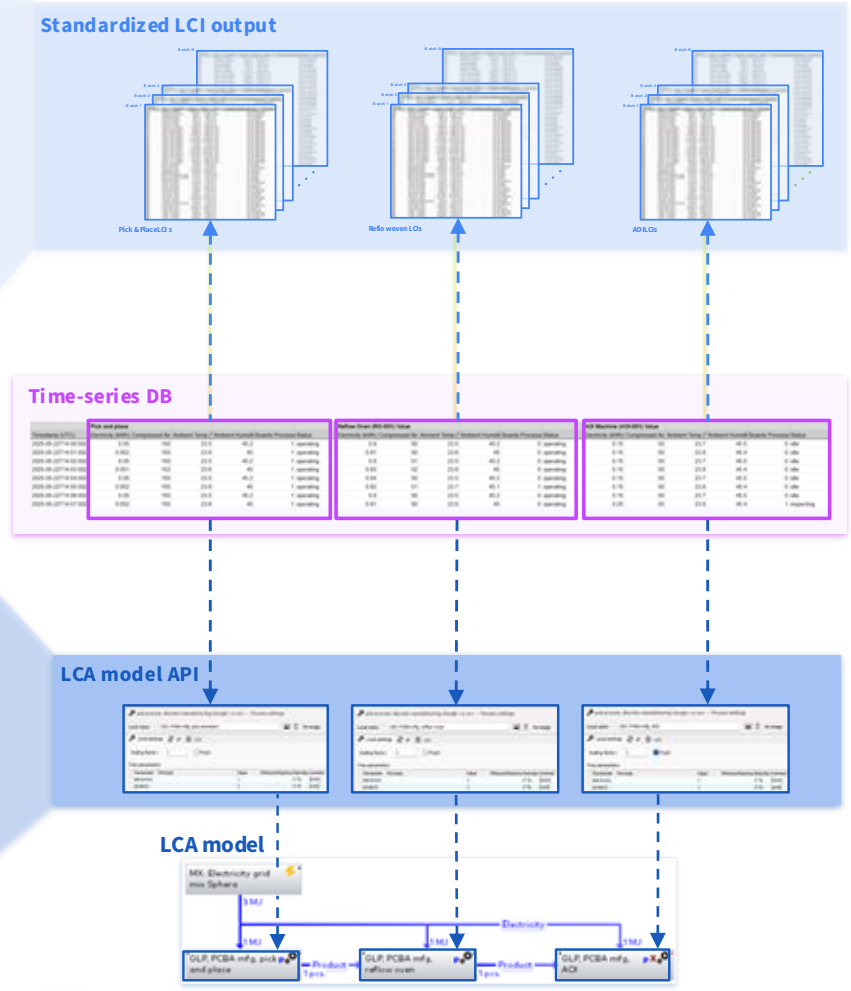
3. Manipulation + orchestration
Support dynamic metadata linking, aggregation, and partitioning.

Approach 1 |
Direct LCI generation

Evaluates real-time boundaries and transforms production telemetry into structured, ISO-compliant datasets instantly.

Approach 2 |
LCA model integration

Connects telemetry data to parameterized LCA models via API endpoints for systematic parameter adjustment without manual rebuilds.



Data use case: Predictive analytics for energy efficiency

GenAI Root Cause Analysis (RCA)

Data ingest

Combines live telemetry JSON streams with PDF machine manuals, maintenance logs, and troubleshooting guides into a Vertex AI RAG corpus

ML modeling

Uses Gemini and machine learning methods to auto-generate plain-language root-cause summaries and step-by-step remediation advice when power anomalies occur

Intended operational and sustainability outcome

Immediate 'Observer Effect' savings: Surfaces phase imbalances and idle equipment draw for zero-capital energy fixes

Reduced response time: Drastically lower Mean Time to Repair (MTTR) for shop-floor technicians

Next steps: Scaling LCA and standardization

Deployment scope

Expand to other manufacturing domains (e.g., mechanicals, PCB):

- **Widen data collection** beyond electricity to include other contributors like compressed air or reflow oven emissions
- **Broaden optimization** to include factors like minimizing manufacturing scrap

Dynamic LCA

Develop technology to automate on-demand LCI generation:

- **On-demand inventories** programmatically compiled from raw factory data rather than relying on static estimates
- **Agentic workflows** to evaluate System-of-Interest boundaries in compliance with ISO standards

Standardization

Establish open-source data schemas and specifications:

- **OCP Workstreams** integration to align shared data schemas (e.g., PCR)
- **Carbon Disclosure** embedding empirical power specifications into the Base Specification

Call to Action

Leverage **digital infrastructure**

We encourage other hardware manufacturers to join us in:

- **Open-source schemas:** Build open, vendor-neutral data schemas connecting telemetry to ILCD-conformant life cycle inventories.
- **Device-level profiling:** Adopt power profiling during hardware manufacturing and testing to replace static estimates across supply chains.

Unified **harmonization**

Our goal is to harmonize this approach with other OCP workstreams (PCR and carbon disclosure) to:

- **Share methodologies:** Develop aligned, empirical calculation methodologies.
- **Unify standards:** Create and agree on a consistent baseline for the industry.
- **Drive adoption:** Push for standard industry-wide implementation and integration.

Thank You!

